

Kenneth Bruskiewicz

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Vancouver, BC, Canada

Software Engineer — Distributed Systems, Scientific Data Infrastructure, Secure Pipelines

Focus: ontology-driven systems, federated data architectures, and high-integrity data workflows for scientific and research environments.

Skills

Database Management

Constructing REST APIs and schemas for NoSQL, graph, and SQL/relational databases like Neo4J, PostgreSQL, and DynamoDB, using OpenAPI.

Domain Translation & Ontology Engineering

Fluent in formal ontology languages (LinkML, RDF/OWL) and experienced working directly with scientific SMEs to model complex domains into implementable architectures and solutions.

DevOps and ETL Pipelines

Deploying custom big data upload and processing pipelines on the cloud using Docker, Terraform, and AWS, handling giga-to-terabyte-sized transfers.

Application Engineering

Implementing web applications that interact with rich data in real-time. Using Vue.js, React, and Node. Augmented by D3, Vega, Matplotlib, and Redash.

Experience

Research Application Developer – 1QBit Information Technologies

Contract, Q2-4 2025

React, Next.js, Python, Auth0

Built and shipped core features for TopQAD™, a quantum architecture design suite developed with UBC for IEEE 2025, funded by MITACS.

- Deployed mTLS and MFA via Auth0, synchronizing session tokens between the TopQAD™ SPA and Python SDK to enable precise API throttling per use-case, eliminating possible peak load failures during IEEE launch.
- Implemented role-based ACL surfacing features based on each user's terms & conditions acceptance state for application compliance.

Research App Developer / Solutions Architect – Simon Fraser University, CIDGOH Lab Q1,3-4 2023, 2024

Node.js, JQuery, YAML (LinkML)

Designed federated data infrastructure for distributed scientific consortia, working at the intersection of ontology engineering, data governance, and research tooling.

- Wrote technical vision and strategy section of grant to Genome Canada, funded for \$4 million CAD with in-kind contributions from Genome BC and other genomics institutions and cloud providers, to \$10 million.
- Architected a proposal for a privacy-first ontology-driven Data Mesh for nine crop-domain scientific teams for Genome Canada's Agrifood initiative, bringing private-/public-sector partners on board for implementation.
- Brought "Data Harmonizer" to product maturity, adding internationalisation support and one-to-many ontological relation support to Data Harmonizer with the LinkML specification language.

Consulting Software Developer – University of Colorado Boulder

Q3-Q4 2022

Node.js, Vue, Biolink Model Ontology (LinkML), Neo4J

- Built a bespoke dashboard for reporting and visualising the compliance of distributed and federated knowledge APIs with the Biolink Model Ontology, an industry-grade biomedical ontology written in the RDF/OWL framework, for the National Consortium of Translational Sciences (NCATS).
- Enabled NCATS to monitor and enforce semantic interoperability across a distributed network of independent biomedical APIs passing representations converted to Neo4J queries – a compliance problem that only exists at the intersection of federated architecture and formal ontology.

Consulting Software Developer – Prometheus Biosciences

2021-2022

Docker, AWS (EC2, S3, Lambda), Terraform, Typescript, Redash, Python

- Built high-throughput data ingestion system handling TB-scale transfers via FTP, reporting upload completion on a Redash dashboard, with notifications sent via email, to eliminate confusion regarding an upload status and compliance with schemas devised along with the Data Science & Engineering team.
- Designed ephemeral credential system for secure upload sessions, Created a serverless auth-token site for issuing upload sessions to guarantee upload security through links that expire upon use. Written in Typescript, with static site templates built and served through AWS Lambda.
- Automated cloud infrastructure provisioning and IAM policy design using Terraform.

Intermediate Front-end Developer – Broad Institute, Flannick Lab[†]

2020-2021

Vue.js, Node.js, Javascript, D3, Vega, React, Javascript, YAML (LinkML)

- Developed a filter component for cross-filtering faceted visualisations across the HuGEAMP research portals for insightful explorations of categorical and quantitative biomedical data using Vue.js.
- Converted a third-party GWAS/gene-annotation track tool (LocusZoom) into a composable, “drop-in” web component placeable on any page of HuGEAMP by other devs, collaborating with the original maintainer.
- Championed integrating third-party API “MyGene.info” to HuGEAMP to contextualise gene information by adding a missing kind of data based on SME feedback on what would bring more value to HuGEAMP.
- Authored a colour scheme generating library for labelling categorical data based on the designer’s style guide to consistently style an unknown number of labels with no colour overlaps.

Software Developer – Oregon Health and Science University[†]

Q2 2019

Python, Common Workflow Language (CWL), Jupyter Notebooks, YAML (LinkML), Neo4J, RDF

- Prototyped a method for converting Jupyter Notebooks into tools that can be composed and run in workflows, based on ontological types in the Biolink Model.

Junior Software Developer – Scripps Institute & Lawrence Berkeley Laboratories[†]

2016

Cytoscape.js, Java (Vaadin 8), Python, Neo4J, FastAPI, RDF/OWL

- Integrated Cytoscape.js as the graph visualisation for “Knowledge.Bio”, a prototype for interacting with semantic networks of biomedical knowledge during the pilot stage of the NCATS initiative of the NIH. This was the first direct application of the Biolink Model, an RDF-based model for biomedical data, to researcher workflows involving the Semantic Web. (Bruskiewich, Huellas-Bruskiewicz et al).
- Aided in designing and applying “Knowledge Beacons,” a FastAPI standard for transferring biomedical graph data using REST. Wrote middleware for use in “Knowledge.Bio” using Spring. (Hannestad, Dančík et al).

[†] Consulting through STAR Informatics/Delphinai Corporation

Education

B.Sci., Interactive Arts & Technology – Simon Fraser University, Vancouver, BC

May 2019

Diploma, Data Science – Brainstation, Vancouver, BC

May 2023-Aug 2023

BCS., Computer Science – University of British Columbia, Vancouver, BC

Q4 2024-est. 2027

Citations

Bruskiewich, R., Huellas-Bruskiewicz, K., Ahmed, F., Kaliyaperumal, R., Thompson, M., Schultes, E., Hettne, K. M., Su, A. I., & Good, B. M. (2016). Knowledge.Bio: A Web Application for Exploring, Building and Sharing Webs of Biomedical Relationships Mined from PubMed. Cold Spring Harbor Laboratory. <https://doi.org/10.1101/055525>

Hannestad, L. M., Dančik, V., Godden, M., Suen, I. W., Huellas-Bruskiewicz, K. C., Good, B. M., Mungall, C. J., & Bruskiewich, R. M. (2020). Knowledge Beacons: Web Service Workflow for FAIR Data Harvesting of Distributed Biomedical Knowledge. Cold Spring Harbor Laboratory. <https://doi.org/10.1101/2020.04.06.027425>

Costanzo, M. C., von Grotthuss, M., Massung, J., Jang, D., Caulkins, L., Koesterer, R., Gilbert, C., Welch, R. P., Kudtarkar, P., Hoang, Q., Boughton, A. P., Singh, P., Sun, Y., Duby, M., Moriondo, A., Nguyen, T., Smadbeck, P., Alexander, B. R., Brandes, M., Huellas-Bruskiewicz, K. C., ... Zaghloul, N. A. (2023). The Type 2 Diabetes Knowledge Portal: An open access genetic resource dedicated to type 2 diabetes and related traits. In *Cell Metabolism* (Vol. 35, Issue 4, pp. 695-710.e6). Elsevier BV. <https://doi.org/10.1016/j.cmet.2023.03.001>